Equal-area projection

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$$x = H\ \cos D \tag{1a}$$
$$y = H\ \sin D \tag{1b}$$
$$z = R \sin I \tag{1c}$$
$$H = \sqrt{x^2 + y^2} = R \cos I \tag{1d}$$
$$R = \sqrt{x^2 + y^2 + z^2} \tag{1e}$$

$$A^{+}_{cap} = \int_{-\pi}^{\pi} \int_{I}^{\pi/2} R^2\ \cos I'\, dI' dD' = 2\pi R^2 (1 - \sin I)\,, \quad I > 0 \tag{2a}$$

$$A^{-}_{cap} = \int_{-\pi}^{\pi} \int_{-\pi/2}^{I} R^2\ \cos I'\, dI' dD' = 2\pi R^2 (1 - \sin I)\,, \quad I < 0 \tag{2b}$$

$$A_{cap} = 2\pi R^2 (1 - \sin |I|) \tag{3}$$

$$A_{circ} = \pi r^2 \tag{4}$$

$$\frac{A_{circ}}{A_{cap}} = \frac{1}{2} \quad \longrightarrow \quad r = R\sqrt{1 - \sin |I|} \tag{5}$$

$$\tilde{r}_{xy} = \sqrt{1 - |z|} \tag{6a}$$
$$x' = \tilde{r}_{xy}\ \cos D = \frac{\tilde{r}_{xy}}{\sqrt{x^2+y^2}}\, x \tag{6b}$$
$$y' = \tilde{r}_{xy}\ \sin D = \frac{\tilde{r}_{xy}}{\sqrt{x^2+y^2}}\, y \tag{6c}$$

$$\tilde{r}_{yz} = \sqrt{1 - |x|} \tag{7a}$$
$$y' = \frac{\tilde{r}_{yz}}{\sqrt{y^2+z^2}}\, y \tag{7b}$$
$$z' = \frac{\tilde{r}_{yz}}{\sqrt{y^2+z^2}}\, z \tag{7c}$$